File Permissions and Altering using the Numbers, Using +,- and Checking

```
vanam@Uday:~/devops$ touch file1.txt file2.txt
vanam@Uday:~/devops$ ls -l file1.txt file2.txt
-rwxr-x--- 1 vanam vanam 0 Oct 12 09:54 file1.txt
-rwxr-xr-x 1 vanam vanam 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ chmod 755 file1.txt
vanam@Uday:~/devops$ ls -l
total 0
-rwxr-xr-x 1 vanam vanam 0 Oct 12 09:54 file1.txt
-rwxr-xr-x 1 vanam vanam 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ chmod 644 file2.txt
vanam@Uday:~/devops$ ls -l
total 0
-rwxr-xr-x 1 vanam vanam 0 Oct 12 09:54 file1.txt
-rw-r--r-- 1 vanam vanam 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ chmod +x file2.txt
vanam@Uday:~/devops$ ls -l
total 0
-rwxr-xr-x 1 vanam vanam 0 Oct 12 09:54 file1.txt
-rwxr-xr-x 1 vanam vanam 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ chmod u+x file1.txt
vanam@Uday:~/devops$ ls -l
total 0
-rwxr-xr-x 1 vanam vanam 0 Oct 12 09:54 file1.txt
-rwxr-xr-x 1 vanam vanam 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ chmod u=rwx,g=rx,o= file1.txt
vanam@Uday:~/devops$ ls -l
total 0
-rwxr-x--- 1 vanam vanam 0 Oct 12 09:54 file1.txt
-rwxr-xr-x 1 vanam vanam 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ chmod ug+rw,o-rwx file1.txt file2.txt
vanam@Uday:~/devops$ ls -l
total 0
```

User Management User addition, group addition and then changing the groups

```
vanam@Uday:~/devops$ chmod ug+rw,o-rwx file1.txt file2.txt
vanam@Uday:~/devops$ ls -l
total 0
-rwxrwx--- 1 vanam vanam 0 Oct 12 09:54 file1.txt
-rwxrwx--- 1 vanam vanam 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ sudo useradd testuser
[sudo] password for vanam:
vanam@Uday:~/devops$ sudo groupadd testgroup
vanam@Uday:~/devops$ sudo usermod -aG testgroup testuser
vanam@Uday:~/devops$ id testuser
uid=1001(testuser) gid=1001(testuser) groups=1001(testuser),1002(testgroup)
vanam@Uday:~/devops$ groups testuser
testuser : testuser testgroup
vanam@Uday:~/devops$ sudo chown testuser file1.txt
vanam@Uday:~/devops$ ls -l
total 0
-rwxrwx--- 1 testuser vanam 0 Oct 12 09:54 file1.txt
-rwxrwx--- 1 vanam    vanam 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ sudo chown testuser:testgroup file2.txt
vanam@Uday:~/devops$ ls -l
\total 0
-rwxrwx--- 1 testuser vanam     0 Oct 12 09:54 file1.txt
-rwxrwx--- 1 testuser testgroup 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ sudo chgrp testgroup file1.txt
vanam@Uday:~/devops$ ls -l
total 0
-rwxrwx--- 1 testuser testgroup 0 Oct 12 09:54 file1.txt
-rwxrwx--- 1 testuser testgroup 0 Oct 12 09:54 file2.txt
vanam@Uday:~/devops$ sudo usermod -L testuser
vanam@Uday:~/devops$ sudo usermod -U testuser
usermod: unlocking the user's password would result in a passwordless account.
You should set a password with usermod -p to unlock this user's password.
```

Locking and Unlocking the User and Set password again and date command

```
vanam@Uday:~/devops$ sudo usermod -L testuser
vanam@Uday:~/devops$ sudo usermod -U testuser
usermod: unlocking the user's password would result in a passwordless account.
You should set a password with usermod -p to unlock this user's password.
vanam@Uday:~/devops$ sudo passwd testuser
New password:
Retype new password:
passwd: password updated successfully
vanam@Uday:~/devops$ date
Mon Oct 12 10:04:17 UTC 2026
vanam@Uday:~/devops$ date +"%Y-%m-%d %H:%M:%S"
2026-10-12 10:04:27
vanam@Uday:~/devops$ date -d "yesterday"
date -d "next week"
date -d "30 days ago"
Sun Oct 11 10:04:42 UTC 2026
Mon Oct 19 10:04:42 UTC 2026
Sat Sep 12 10:04:42 UTC 2026
```

Switching the user and exit

```
vanam@Uday:~/devops$ su - testuser
Password:
su: warning: cannot change directory to /home/testuser: No such file or directory
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
Unable to setup logging. [Errno 2] No such file or directory: '/home/testuser/.landscape'

 System information as of Mon Oct 12 10:10:44 UTC 2026

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

This message is shown once a day. To disable it please create the
/home/testuser/.hushlogin file.
touch: cannot touch '/home/testuser/.motd_shown': No such file or directory
$  exit
vanam@Uday:~/devops$
```

Installing the Cal And Viewing the Cal for the month and Year

```
vanam@Uday:~/devops$ cal
Command 'cal' not found, but can be installed with:
sudo apt install ncal
vanam@Uday:~/devops$ sudo apt install ncal
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  ncal
0 upgraded, 1 newly installed, 0 to remove and 62 not upgraded.
Need to get 21.0 kB of archives.
After this operation, 59.4 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 ncal amd64 12.1.8 [21.0 kB]
Fetched 21.0 kB in 1s (26.9 kB/s)
Selecting previously unselected package ncal.
(Reading database ... 48453 files and directories currently installed.)
Preparing to unpack .../archives/ncal_12.1.8_amd64.deb ...
Unpacking ncal (12.1.8) ...
Setting up ncal (12.1.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
vanam@Uday:~/devops$ cal
    October 2026
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31
```

```
vanam@Uday:~/devops$ cal 10 2026
    October 2026
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

vanam@Uday:~/devops$ cal 2026
                               2026
      January               February                March
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
             1  2  3   1  2  3  4  5  6  7   1  2  3  4  5  6  7
 4  5  6  7  8  9 10   8  9 10 11 12 13 14   8  9 10 11 12 13 14
11 12 13 14 15 16 17  15 16 17 18 19 20 21  15 16 17 18 19 20 21
18 19 20 21 22 23 24  22 23 24 25 26 27 28  22 23 24 25 26 27 28
25 26 27 28 29 30 31                        29 30 31


       April                   May                   June
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
          1  2  3  4                  1  2      1  2  3  4  5  6
 5  6  7  8  9 10 11   3  4  5  6  7  8  9   7  8  9 10 11 12 13
12 13 14 15 16 17 18  10 11 12 13 14 15 16  14 15 16 17 18 19 20
19 20 21 22 23 24 25  17 18 19 20 21 22 23  21 22 23 24 25 26 27
26 27 28 29 30        24 25 26 27 28 29 30  28 29 30
                      31

        July                  August              September
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
```

Text tricks in Ubuntu

reversing

```
vanam@Uday:~$ echo "hello linux" | rev
xunil olleh
vanam@Uday:~$
```

Shuffling

```
vanam@Uday:~$ shuf -e tea coffee juice -n 1
coffee
vanam@Uday:~$ shuf -e tea coffee juice -n 2
tea
coffee
vanam@Uday:~$ shuf -e tea coffee juice watermelon apple -n 2
watermelon
apple
vanam@Uday:~$
```

Converting Uppercase to Lowercase

```
vanam@Uday:~$ echo "LINUX USER" | tr 'A-Z' 'a-z'
linux user
```

Factors

```
vanam@Uday:~$ factor 120
120: 2 2 2 3 5
vanam@Uday:~$ factor 20
20: 2 2 5
```

Sequence

```
vanam@Uday:~$ seq 1 5
1
2
3
4
5
```

Printing Statistics

```
vanam@Uday:~/devops$ stat file1.txt
  File: file1.txt
  Size: 0               Blocks: 0          IO Block: 4096   regular empty file
Device: 8,48    Inode: 1710        Links: 1
Access: (0770/-rwxrwx---)  Uid: ( 1001/testuser)   Gid: ( 1002/testgroup)
Access: 2026-10-12 09:54:43.661286790 +0000
Modify: 2026-10-12 09:54:43.661286790 +0000
Change: 2026-10-12 10:00:24.153262709 +0000
 Birth: 2026-10-12 09:53:18.740565794 +0000
```